

Experiment 5: Docker - Volumes, Environment Variables, Monitoring & Networks

Part 1: Docker Volumes - Persistent Data Storage

Lab 1: Understanding Data Persistence

The Problem: Container Data is Ephemeral

```
docker run -it --name test-container ubuntu /bin/bash
echo "Hello World" > message.txt
cat message.txt
exit
docker stop test-container
docker rm test-container
docker run -it --name test-container ubuntu /bin/bash
cat message.txt
# cat: message.txt: No such file or directory
```

```
# Anonymous volume
docker run -d -v /app/data --name web1 nginx
docker volume ls
docker inspect web1 | grep -A 5 Mounts
```

```

abhinav_pandey@LUCIF3R:~$ docker run -it --name test-container ubuntu /bin/bash
root@97f6e9b24dd8:/# echo "Hello World" > /data/message.txt
bash: /data/message.txt: No such file or directory
root@97f6e9b24dd8:/# cat /data/message.txt
cat: /data/message.txt: No such file or directory
root@97f6e9b24dd8:/# mkdir data
root@97f6e9b24dd8:/# echo "Hello World" > /data/message.txt
root@97f6e9b24dd8:/# cat /data/message.txt
Hello World
root@97f6e9b24dd8:/# exit
exit
abhinav_pandey@LUCIF3R:~$ docker start test-container
test-container
abhinav_pandey@LUCIF3R:~$ docker exec test-container cat /data/message.txt
Hello World
abhinav_pandey@LUCIF3R:~$ docker stop test-container
test-container
abhinav_pandey@LUCIF3R:~$ docker rm test-container
test-container
abhinav_pandey@LUCIF3R:~$ docker run -it --name test-container ubuntu /bin/bash
root@ecbd2981748c:/# cat message.txt
cat: message.txt: No such file or directory
root@ecbd2981748c:/# exit
exit
abhinav_pandey@LUCIF3R:~$ docker run -d -v /app/data --name web1 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
448ea5cac5d5: Pull complete
4a038fd18db1: Pull complete
88d1d984b765: Pull complete
5435b2dcdf5c: Pull complete
84e114c2bb36: Pull complete
054715a6bffa: Pull complete
7b5d674621c2: Pull complete
cc0cf959117b: Download complete
3eff8a97d435: Download complete
Digest: sha256:7f0adca1fc6c29c8dc49a2e90837a10ba20dc266baed0988e9fb4d0d8b85ba0
Status: Downloaded newer image for nginx:latest
85ef4033cc3b5afa9dfaed7fa2426c4ab1959fdef3e0ccf06f582da5ee88957e
abhinav_pandey@LUCIF3R:~$ docker volume ls
DRIVER      VOLUME NAME
local       441ed787faa660c56d13c08ea1becaa8729a414d1c409eff372dd7baf18d6e62
abhinav_pandey@LUCIF3R:~$ docker inspect web1 | grep -A 5 Mounts
"Mounts": [
  {
    "Type": "volume",
    "Name": "441ed787faa660c56d13c08ea1becaa8729a414d1c409eff372dd7baf18d6e62",

```

Lab 2: Volume Types

1. Named Volumes

```

docker volume create mydata
docker run -d -v mydata:/app/data --name web2 nginx
docker volume ls
docker volume inspect mydata
docker run -d -v ~/myapp-data:/app/data --name web3 nginx

```

```

abhinav_pandey@LUCIF3R:~$ docker inspect web1 | grep -A 5 Mounts
"Mounts": [
  {
    "Type": "volume",
    "Name": "441ed787faa660c56d13c08ea1becaa8729a414d1c409eff372dd7baf18d6e62",
    "Source": "/var/lib/docker/volumes/441ed787faa660c56d13c08ea1becaa8729a414d1c409eff372dd7baf18d6e62/_data",
    "Destination": "/app/data",
abhinav_pandey@LUCIF3R:~$ docker volume create mydata
mydata
abhinav_pandey@LUCIF3R:~$ docker run -d -v mydata:/app/data --name web2 nginx
b95ddf6d421529fd06bca68535e910364f152ccabbe43c84c00b64151078b1bb
abhinav_pandey@LUCIF3R:~$ docker volume ls
DRIVER      VOLUME NAME
local       441ed787faa660c56d13c08ea1becaa8729a414d1c409eff372dd7baf18d6e62
local       mydata
abhinav_pandey@LUCIF3R:~$ docker volume inspect mydata
[
  {
    "CreatedAt": "2026-04-09T13:44:05Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/mydata/_data",
    "Name": "mydata",
    "Options": null,
    "Scope": "local"
  }
]
abhinav_pandey@LUCIF3R:~$ docker run -d -v ~/myapp-data:/app/data --name web3 nginx
78e9t69261c6dbed1d1c7e261ddd0t+d5ceb6fce4c5ee1b38fe3f5da27b09b

```

2. Bind Mounts (Host Directory)

```
mkdir myapp-data
ls
docker run -d -v $(pwd)/myapp-data:/app/data --name web3 nginx
echo "From Host" > myapp-data/host-file.txt
docker exec -it web3 bash
ls /app/data
cat /app/data/host-file.txt
# From Host
exit
docker run -d \
  --name mysql-db \
  -v mysql-data:/var/lib/mysql \
```

```

abhinav_pandey@LUCIF3R:~$ mkdir myapp-data
abhinav_pandey@LUCIF3R:~$ ls
myapp-data
abhinav_pandey@LUCIF3R:~$ docker run -d -v $(pwd)/myapp-data:/app/data --name web3 nginx
d728b47aab0f4b4bfcc4e97b427d7db289b94c8b6dcc59ba92d77374f9dbdc4926
abhinav_pandey@LUCIF3R:~$ echo "From Host" > myapp-data/host-file.txt
abhinav_pandey@LUCIF3R:~$ docker exec -it web3 bash
root@d728b47aab0f:/# ls /app/data
host-file.txt
root@d728b47aab0f:/# cat /app/data/host-file.tx
cat: /app/data/host-file.tx: No such file or directory
root@d728b47aab0f:/# cat /app/data/host-file.txt
From Host
root@d728b47aab0f:/# exit
exit
abhinav_pandey@LUCIF3R:~$ docker run -d \
--name mysql-db \
-v mysql-data:/var/lib/mysql \

```

Lab 3: Practical Volume Examples

Example 1: Database with Persistent Storage

```
# mysql:8.0 pulling layers
docker stop mysql-db
docker rm mysql-db
docker run -d \
  --name new-mysql \
  -v mysql-data:/var/lib/mysql \
  -e MYSQL_ROOT_PASSWORD=secret \
  mysql:8.0
mkdir ~/nginx-config
echo 'server {
    listen 80;
    server_name localhost;
    location / {
        return 200 "Hello from mounted config!";
    }
}' > ~/nginx-config/nginx.conf
docker run -d \
  --name nginx-custom \
  -p 8080:80 \
  -v ~/nginx-config/nginx.conf:/etc/nginx/conf.d/default.conf \
  nginx
```

```
abhinav_pandey@LUCIF3R:~$ docker stop mysql-db
mysql-db
docker rm mysql-db
mysql-db
docker run -d \
  --name new-mysql \
  -v mysql-data:/var/lib/mysql \
  -e MYSQL_ROOT_PASSWORD=secret \
  mysql:8.0
mysql:8.0
Error response from daemon: No such container: mysql-db
Error response from daemon: No such container: mysql-db
9dd09592720c041f0fb8f099f5337b8a5b34887099509d9abccfd29cfc04bd11
abhinav_pandey@LUCIF3R:~$ mkdir ~/nginx-config
mkdir: cannot create directory '/home/abhinav_pandey/nginx-config': File exists
abhinav_pandey@LUCIF3R:~$ mkdir nginx-config
mkdir: cannot create directory 'nginx-config': File exists
abhinav_pandey@LUCIF3R:~$ echo 'server {
    listen 80;
    server_name localhost;
    location / {
        return 200 "Hello from mounted config!";
    }
}' > ~/nginx-config/nginx.conf
server {
    listen 80;
    server_name localhost;
    location / {
        return 200 "Hello from mounted config!";
    }
}
abhinav_pandey@LUCIF3R:~$ echo 'server {
    listen 80;
    server_name localhost;
    location / {
        return 200 "Hello from mounted config!";
    }
}' > ~/nginx-config/nginx.conf
abhinav_pandey@LUCIF3R:~$ docker run -d \
  --name nginx-custom \
  -p 8080:80 \
  -v ~/nginx-config/nginx.conf:/etc/nginx/conf.d/default.conf \
  nginx
44252783d510a2ce1fb9e66cc22ae7e2746e22a046e4fc8192bf2ad2d26e7c07
```

Example 2: Web App with Configuration Files

```
curl http://localhost:8080
# Hello from mounted config!
curl http://localhost:8080
# Hello from mounted config!
docker volume create app-volume
docker volume inspect app-volume
docker volume prune
# Deleted Volumes listed
# Total reclaimed space: 246.1MB
docker run -d \
  --name app1 \
  -e DATABASE_URL="postgres://user:pass@db:5432/mydb" \
  -e DEBUG="true" \
  -p 3000:3000 \
  nginx
# Error: container name "/app1" already in use
```

```
docker rm mysql-db
docker run -d \
  --name new-mysql \
  -v mysql-data:/var/lib/mysql \
  -e MYSQL_ROOT_PASSWORD=secret \
  mysql:8.0
Error response from daemon: No such container: mysql-db
Error response from daemon: No such container: mysql-db
9dd0592720c041f9fb8f099f5337b8a5b34887099509d9abccfd29cfc04bd11
abhinav_pandey@UKIF3R:~$ mkdir -p /nginx-conf
mkdir: cannot create directory '/home/abhinav_pandey/nginx-conf': File exists
abhinav_pandey@UKIF3R:~$ mkdir nginx-conf
mkdir: cannot create directory 'nginx-conf': File exists
abhinav_pandey@UKIF3R:~$ echo 'server {
    listen 80;
    server_name localhost;
    location / {
        return 200 "Hello from mounted config!";
    }
}' > /nginx-conf/nginx.conf
abhinav_pandey@UKIF3R:~$ docker run -d \
  --name nginx-custom \
  -p 8080:80 \
  -v /nginx-conf/nginx.conf:/etc/nginx/conf.d/default.conf \
  nginx
0422783d510a2celfb9e66cc22ae7e2746e22a046e4fc0192bf2ad2d26e7c07
abhinav_pandey@UKIF3R:~$ curl http://localhost:8080
Hello from mounted config!
docker volume create app-volume
abhinav_pandey@UKIF3R:~$ docker volume inspect app-volume
[
  {
    "CreatedAt": "2026-04-09T13:56:35Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/app-volume/_data",
    "Name": "app-volume",
    "Options": null,
    "Scope": "local"
  }
]
abhinav_pandey@UKIF3R:~$ docker volume prune
WARNING! This will remove anonymous local volumes not used by at least one container.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B
abhinav_pandey@UKIF3R:~$ docker run -d \
  --name app1 \
  -e DATABASE_URL="postgres://user:pass@db:5432/mydb" \
  -e DEBUG="true" \
  -p 3000:3000 \
  nginx
05c915b7dab5e47275c9c724630ac1d9816b408091b8f7834970c4ff0a403c05
```

Part 2: Environment Variables

Lab 1: Setting Environment Variables

Method 1: Using -e flag

```
docker rm -f app11
docker run -d \
  --name app11 \
  -e DATABASE_URL="postgres://user:pass@db:5432/mydb" \
  -e DEBUG="true" \
  -p 3005:3000 \
  nginx
docker ps
# Shows: app11, nginx-custom, new-mysql, web3, web2, web1, node-container
```

```
abhinav.pandey@LUCIF3R: ~$ docker rm -f app11
docker run -d \
  --name app11 \
  -e DATABASE_URL="postgres://user:pass@db:5432/mydb" \
  -e DEBUG="true" \
  -p 3005:3000 \
  nginx
Error response from daemon: No such container: app11
f937dff898130600ba9692043d51d97691c34684eee6aa48fb8d466ffdc3100
abhinav.pandey@LUCIF3R: ~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
f937dff89813	nginx	"/docker-entrypoint..."	9 seconds ago	Up 9 seconds	80/tcp, 0.0.0.0:3005->3000/tcp, [::]:3005->3000/tcp	app11
45c915b7dacb	nginx	"/docker-entrypoint..."	4 minutes ago	Up 4 minutes	80/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	app1
d4252783d510	nginx	"/docker-entrypoint..."	6 minutes ago	Up 6 minutes	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	nginx-custom
9dd09592720c	mysql:8.0	"/docker-entrypoint.s..."	7 minutes ago	Up 7 minutes	3306/tcp, 33060/tcp	new-mysql
d728b47aab0f	nginx	"/docker-entrypoint..."	14 minutes ago	Up 14 minutes	80/tcp	web3
b95ddf64215	nginx	"/docker-entrypoint..."	18 minutes ago	Up 18 minutes	80/tcp	web2
85ef4033cc3b	nginx	"/docker-entrypoint..."	19 minutes ago	Up 19 minutes	80/tcp	web1

```
docker exec -it app11 bash
env | grep DATABASE
# DATABASE_URL=postgres://user:pass@db:5432/mydb
env | grep DEBUG
# DEBUG=true
exit
docker rm -f app11
```

Method 2: Using --env-file

```
mkdir env-lab
cd env-lab
echo "DATABASE_HOST=localhost" > .env
echo "DATABASE_PORT=5432" >> .env
echo "API_KEY=secret123" >> .env
cat .env
# DATABASE_HOST=localhost
# DATABASE_PORT=5432
# API_KEY=secret123
docker run -d --name app2 --env-file .env nginx
# Error: name "/app2" already in use
docker run -d --name app22 --env-file .env nginx
docker exec -it app22 bash
echo $DATABASE_HOST
```

```
abhinav_pandey@LUCIF3R: ~$ docker exec -it app11 bash
env | grep DATABASE
root@f937dff89813:/# env | grep DEBUG
DEBUG=true
root@f937dff89813:/# exit
exit
abhinav_pandey@LUCIF3R: ~$ docker rm -f app11
app11
```

```
# localhost
echo $DATABASE_PORT
# 5432
echo $API_KEY
# secret123
exit
docker rm -f app22
cd ..
mkdir flask-env-app
cd flask-env-app
echo "flask" > requirements.txt
nano app.py
nano Dockerfile
docker build -t flask-env-app .
# [+] Building 7.4s (10/10) FINISHED
```

[illegible]

Lab 2: Flask App with Environment Variables

Dockerfile Build and Run

```
# Build output - all 5 steps completing successfully
docker run -d \
  --name flask1 \
  -p 5001:5000 \
  -e DATABASE_HOST=mydb \
  -e DEBUG=true \
  -e API_KEY=supersecret \
  flask-env-app
# Error: port 0.0.0.0:5000 already allocated
docker rm -f flask1
docker run -d \
  --name flask1 \
  -p 5001:5000 \
  -e DATABASE_HOST=mydb \
  -e DEBUG=true \
  -e API_KEY=supersecret \
  flask-env-app
# Container started successfully
```

```
0.1s
abhinav_pandey@LUCIF3R:~/flask-env-app$ docker run
--name flask1 \
-p 5001:5000 \
-e DATABASE_HOST=mydb \
-e DEBUG=true \
-e API_KEY=supersecret \
flask-env-app
39ea8df64827cd1bb68e38da57f18044590bd19b69605bd2a26c6a8db09000c3
```

Lab 3: Test Environment Variables

```
# Browser: localhost:5001/config
# Output:
# {
#   "db_host": "mydb",
#   "debug": true,
#   "has_api_key": true
# }
```

```

abhinav_pandey@LUCIF3R:~/flask-env-apps docker rm -f flask1

docker run -d \
  --name flask1 \
  -p 5001:3000 \
  -e DATABASE_HOST=mydb \
  -e DEBUG=true \
  -e API_KEY=supersecret \
  flask-env-app
Error response from daemon: No such container: flask1
8521cbe18941a217b3b90fee3da637e32c8e4154631552150463377b2dedf8c7
abhinav_pandey@LUCIF3R:~/flask-env-apps http://localhost:5001/config
{
  "db_host": "mydb",
  "debug": true,
  "has_api_key": true
}

```

Part 3: Docker Monitoring

Lab 1: `docker stats` - Real-time Container Metrics

```

docker stats
# Live output:
#CONTAINER ID   NAME      CPU %       MEM USAGE / LIMIT   MEM %      NET I/O     BLOCK I/O
#8521cbe18941   flask1    0.28%       40.43MiB / 7.615GiB  0.52%      1.5kB / 777B 0B / 25
#45c915b7dabc   app1      0.00%       18.69MiB / 7.615GiB  0.24%      1.96kB / 126B 172kB /
#d4252783d510   nginx-custom 0.00%       19.02MiB / 7.615GiB  0.24%      3.21kB / 1.21kB 3.91MB
#9dd09592720c   new-mysql 1.04%       349.2MiB / 7.615GiB  4.48%      2.25kB / 126B  36.1MB
#d728b47aab0f   web3      0.00%       18.82MiB / 7.615GiB  0.24%      2.57kB / 126B  0B / 16
#b95ddf64215    web2      0.00%       18.88MiB / 7.615GiB  0.24%      2.76kB / 126B  0B / 12
#85ef4033cc3b   web1      0.00%       18.62MiB / 7.615GiB  0.24%      3.17kB / 126B  0B / 12

```

```

=> [2/4] WORKDIR /app          0.1s
=> [3/4] COPY . .              0.0s
=> [4/4] RUN pip install flask 2.7s
=> exporting to image          0.5s
=> => exporting layers          0.4s
=> => exporting manifest sha256:f617f196faa171010964fe323c2976066df590ef184aa43a90d614bf86be04c8 0.8s
CONTAINER ID   NAME      CPU %       MEM USAGE / LIMIT   MEM %      NET I/O     BLOCK I/O     PID
8521cbe18941   flask1    0.25%       40.43MiB / 7.615GiB  0.52%      1.43kB / 777B  0B / 250kB    3
45c915b7dabc   app1      0.00%       18.69MiB / 7.615GiB  0.24%      1.96kB / 126B  172kB / 12.3kB 25
d4252783d510   nginx-custom 0.00%       19.02MiB / 7.615GiB  0.24%      3.21kB / 1.21kB 3.91MB / 4.1kB 25
9dd09592720c   new-mysql 1.49%       349.2MiB / 7.615GiB  4.48%      2.25kB / 126B  36.1MB / 15.1MB 37
d728b47aab0f   web3      0.00%       18.82MiB / 7.615GiB  0.24%      2.57kB / 126B  0B / 16.4kB    25
b95ddf64215    web2      0.00%       18.88MiB / 7.615GiB  0.24%      2.76kB / 126B  0B / 12.3kB    25
85ef4033cc3b   web1      0.00%       18.62MiB / 7.615GiB  0.24%      3.17kB / 126B  0B / 12.3kB    25

```

```

# docker stats --no-stream (all values show --)
docker stats --format json --no-stream
# JSON output per container with BlockIO, CPUPerc, Container, ID,
# MemPerc, MemUsage, Name, NetIO, PIDs fields
docker stats --no-stream --no-trunc

```

```

abhinav_pandey@LUCIF3R:~/flask-env-app$ docker stats --format json --no-stream
{"BlockIO": {"B": 0, "IOPS": 0, "CPUPerc": "0.18%", "Container": "8521cbe18941a217b3b9fee3da637e32c8e4154631552150463377b2dedf8c7", "ID": "8521cbe18941", "MemPerc": "0.52%", "MemUsage": "40.43MiB / 7.615GiB", "Name": "flask1", "NetIO": "1.5kB / 777B", "PIDs": "3"}, {"BlockIO": {"B": 172kB / 12.3kB, "CPUPerc": "0.00%", "Container": "45c915b7dabc5e47275c9c724630ac1d9816b400891b8f7834970c7f8a483c85", "ID": "45c915b7dabc", "MemPerc": "0.24%", "MemUsage": "18.69MiB / 7.615GiB", "Name": "app1", "NetIO": "1.96kB / 126B", "PIDs": "25"}, {"BlockIO": {"B": 3.91MB / 4.1kB, "CPUPerc": "0.00%", "Container": "d4252783d510a2ce1fb9e66cc22ae7e2746e22a046e4fc8192bf2ad2d26e7c07", "ID": "d4252783d510", "MemPerc": "0.24%", "MemUsage": "19.02MiB / 7.615GiB", "Name": "nginx-custom", "NetIO": "3.21kB / 1.21kB", "PIDs": "25"}, {"BlockIO": {"B": 36.1MB / 15.1MB, "CPUPerc": "1.11%", "Container": "9dd09592728c041f9fb8f099f5337b8a5b34887099509d9abccfd29cfc04bd11", "ID": "9dd09592728c", "MemPerc": "4.48%", "MemUsage": "349.2MiB / 7.615GiB", "Name": "new-mysql", "NetIO": "2.25kB / 126B", "PIDs": "37"}, {"BlockIO": {"B": 0B / 16.4kB, "CPUPerc": "0.00%", "Container": "d728b47aab0fb4bfc4e97b427d7db289b94c8b6dcc59ba92d77374f9dbc4926", "ID": "d728b47aab0f", "MemPerc": "0.24%", "MemUsage": "18.82MiB / 7.615GiB", "Name": "web3", "NetIO": "2.57kB / 126B", "PIDs": "25"}, {"BlockIO": {"B": 0B / 12.3kB, "CPUPerc": "0.00%", "Container": "b95ddf6d6421529fd86bca68535e910364f152ccabbe43c84c00b64151078b1bb", "ID": "b95ddf6d64215", "MemPerc": "0.24%", "MemUsage": "18.88MiB / 7.615GiB", "Name": "web2", "NetIO": "2.76kB / 126B", "PIDs": "25"}, {"BlockIO": {"B": 0B / 12.3kB, "CPUPerc": "0.00%", "Container": "85ef4033cc3b5afa9dfaef7fa2426c4ab1959dfef3e0ccf06f582da5ee88957e", "ID": "85ef4033cc3b", "MemPerc": "0.24%", "MemUsage": "18.62MiB / 7.615GiB", "Name": "web1", "NetIO": "3.17kB / 126B", "PIDs": "25"}
abhinav_pandey@LUCIF3R:~/flask-env-app$ docker stats --no-trunc
CONTAINER ID        NAME               CPU %               MEM USAGE / LIMIT   MEM %               NET I/O             BLOCK I/O           PIDS
8521cbe18941a217b3b9fee3da637e32c8e4154631552150463377b2dedf8c7    flask1             0.17%              40.43MiB / 7.615GiB  0.52%              1.5kB / 777B        0B / 250B           3
45c915b7dabc5e47275c9c724630ac1d9816b400891b8f7834970c7f8a483c85    app1               0.00%              18.69MiB / 7.615GiB  0.24%              1.96kB / 126B       172kB / 12.3kB     25
d4252783d510a2ce1fb9e66cc22ae7e2746e22a046e4fc8192bf2ad2d26e7c07    nginx-custom       0.00%              19.02MiB / 7.615GiB  0.24%              3.21kB / 1.21kB     3.91MB / 4.1kB     25
9dd09592728c041f9fb8f099f5337b8a5b34887099509d9abccfd29cfc04bd11    new-mysql          1.16%              349.2MiB / 7.615GiB  4.48%              2.25kB / 126B       36.1MB / 15.1MB    37
d728b47aab0fb4bfc4e97b427d7db289b94c8b6dcc59ba92d77374f9dbc4926    web3               0.00%              18.82MiB / 7.615GiB  0.24%              2.57kB / 126B       0B / 16.4kB        25
b95ddf6d6421529fd86bca68535e910364f152ccabbe43c84c00b64151078b1bb    web2               0.00%              18.88MiB / 7.615GiB  0.24%              2.76kB / 126B       0B / 12.3kB        25
85ef4033cc3b5afa9dfaef7fa2426c4ab1959dfef3e0ccf06f582da5ee88957e    web1               0.00%              18.62MiB / 7.615GiB  0.24%              3.17kB / 126B       0B / 12.3kB        25

```

Lab 2: docker top - Process Monitoring

```
docker top flask1
```

#UID	PID	PPID	C	STIME
#root	9434	9410	0	14:19
#root	9522	9434	0	14:19

```
#docker top flask1 -ef
#ps aux | grep docker
```

```

abhinav_pandey@LUCIF3R:~/flask-env-app$ docker top flask1
UID PID PPID C STIME TTY TIME CMD
root 9434 9410 0 14:19 ? 00:00:00 python app.py
root 9434 9410 0 14:19 ? 00:00:00 /usr/local/bin/python /app/app.py
abhinav_challa@LUCIF3R:~/flask-env-app$ docker top flask1 -ef
UID PID PPID C STIME TTY TIME CMD
root 9434 9410 0 14:20 ? 00:00:00 python app.py
root 9434 9410 0 14:20 ? 00:00:00 /usr/local/bin/python /app/app.py
abhinav_challa@LUCIF3R:~/flask-env-app$ ps -ef | grep docker
root 1009 0.0 0.0 14:20:10 111960 ? Ssl 13:14 0:28 /usr/bin/docker -H fd:// --containerd=/run/containerd/containerd.sock
root 7937 0.0 0.0 1747108 4992 ? Sl 13:58 0:00 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 8080 -container-ip 172.17.0.6 -container-port 80 -use-listen
-fd
root 7944 0.0 0.0 1747108 4992 ? Sl 13:58 0:00 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 8080 -container-ip 172.17.0.6 -container-port 80 -use-listen-fd
root 8160 0.0 0.0 1747108 4992 ? Sl 13:59 0:00 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 3080 -container-ip 172.17.0.7 -container-port 3080 -use-list
en-fd
root 8166 0.0 0.0 1747108 4880 ? Sl 13:59 0:00 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 3080 -container-ip 172.17.0.7 -container-port 3080 -use-listen-fd
root 9499 0.0 0.0 1746852 4880 ? Sl 14:20 0:00 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 5001 -container-ip 172.17.0.8 -container-port 3080 -use-list
en-fd
root 9586 0.0 0.0 1886448 4992 ? Sl 14:20 0:00 /usr/bin/docker-proxy -proto tcp -host-ip :: -host-port 5001 -container-ip 172.17.0.8 -container-port 3080 -use-listen-fd
abhinav 9700 0.0 0.0 4092 1728 pts/0 S+ 14:26 0:00 grep --color=auto docker

```

```

# docker top monitor-test -ef continued
# showing same worker process list with full command lines
# UID, PID, PPID, C, STIME, TTY, TIME, CMD columns
# worker processes 3998 - 4004

```

Lab 3: docker logs - Application Logs

```

# docker top -ef continued (worker processes 4002-4004)
ps aux | grep docker
# root 615 0.0 0.3 docker-desktop proxy --distro-name Ubuntu-22.04
# aswani 2149 0.0 0.0 grep --color=auto docker
docker logs flask1
# /docker-entrypoint.sh: configuration complete; ready for start up
# nginx/1.29.5 startup notices
# worker processes starting (30-45)

```

```
# docker logs continued:
# nginx worker processes (30-45) started
# HTTP access logs:
# 172.17.0.1 - [02/Mar/2026] "GET / HTTP/1.1" 200 615
# 172.17.0.1 - [02/Mar/2026] "GET /favicon.ico HTTP/1.1" 404 555
# [error] favicon.ico failed (2: No such file or directory)
docker logs --tail 100 flask1
```

```
docker logs -t flask1
# Logs with nanosecond timestamps:
# 2026-03-02T14:30:43.355838305Z /docker-entrypoint.sh: ...
# 2026-03-02T14:30:43.431310396Z [notice] nginx/1.29.5
# 2026-03-02T14:30:43.431372134Z [notice] built by gcc 14.2.0
# (all worker processes with full timestamps)
```

```
docker logs --since 5m flask1
# Shows all logs from last 5 minutes including startup + HTTP requests

docker logs -f --tail 50 -t flask1
# Combined: follow + last 50 lines + timestamps
```

```
abhinav_charla@LUCIF3R:~/flask env-app$ docker logs flask1
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:3000
* Running on http://172.17.0.8:3000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 111-527-213
172.17.0.1 - - [09/Apr/2026 14:19:40] "GET /config HTTP/1.1" 200 -
abhinav_charla@LUCIF3R:~/flask env-app$ docker logs -f flask1
Error response from daemon: no such container: flask1
abhinav_charla@LUCIF3R:~/flask env-app$ docker logs -f flask1
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:3000
* Running on http://172.17.0.8:3000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 111-527-213
172.17.0.1 - - [09/Apr/2026 14:19:40] "GET /config HTTP/1.1" 200 -

^C
abhinav_charla@LUCIF3R:~/flask env-app$ docker logs --tail 100 container-name
Error response from daemon: No such container: container-name
abhinav_charla@LUCIF3R:~/flask env-app$ docker logs --tail 100 flask1
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:3000
* Running on http://172.17.0.8:3000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 111-527-213
172.17.0.1 - - [09/Apr/2026 14:19:40] "GET /config HTTP/1.1" 200 -
abhinav_charla@LUCIF3R:~/flask env-app$ docker logs -f flask1
2026-04-09T14:19:34.422881877Z * Serving Flask app 'app'
2026-04-09T14:19:34.422913444Z * Debug mode: on
2026-04-09T14:19:34.439967389Z WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
2026-04-09T14:19:34.440006670Z * Running on all addresses (0.0.0.0)
2026-04-09T14:19:34.440012751Z * Running on http://127.0.0.1:3000
2026-04-09T14:19:34.440013941Z * Running on http://172.17.0.8:3000
2026-04-09T14:19:34.440014882Z Press CTRL+C to quit
2026-04-09T14:19:34.440756198Z * Restarting with stat
2026-04-09T14:19:34.531377712Z * Debugger is active!
2026-04-09T14:19:34.531597665Z * Debugger PIN: 111-527-213
abhinav_charla@LUCIF3R:~/flask env-app$ docker logs --since 2024-01-15 flask1
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:3000
* Running on http://172.17.0.8:3000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 111-527-213
172.17.0.1 - - [09/Apr/2026 14:19:40] "GET /config HTTP/1.1" 200 -
abhinav_charla@LUCIF3R:~/flask env-app$ docker logs -f --tail 50 -t flask1
2026-04-09T14:19:34.422881877Z * Serving Flask app 'app'
2026-04-09T14:19:34.422913444Z * Debug mode: on
2026-04-09T14:19:34.439967389Z WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
2026-04-09T14:19:34.440006670Z * Running on all addresses (0.0.0.0)
2026-04-09T14:19:34.440012751Z * Running on http://127.0.0.1:3000
2026-04-09T14:19:34.440013941Z * Running on http://172.17.0.8:3000
2026-04-09T14:19:34.440014882Z Press CTRL+C to quit
2026-04-09T14:19:34.440756198Z * Restarting with stat
2026-04-09T14:19:34.531377712Z * Debugger is active!
2026-04-09T14:19:34.531597665Z * Debugger PIN: 111-527-213
2026-04-09T14:19:40.767577014Z 172.17.0.1 - - [09/Apr/2026 14:19:40] "GET /config HTTP/1.1" 200 -
```

```
abhinav_charla@LUCIF3R:~/flask env-app$ docker logs -f --tail 50 -t flask1
2026-04-09T14:19:34.422881877Z * Serving Flask app 'app'
2026-04-09T14:19:34.422913444Z * Debug mode: on
2026-04-09T14:19:34.439967389Z WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
2026-04-09T14:19:34.440006670Z * Running on all addresses (0.0.0.0)
2026-04-09T14:19:34.440012751Z * Running on http://127.0.0.1:3000
2026-04-09T14:19:34.440013941Z * Running on http://172.17.0.8:3000
2026-04-09T14:19:34.440014882Z Press CTRL+C to quit
2026-04-09T14:19:34.440756198Z * Restarting with stat
2026-04-09T14:19:34.531377712Z * Debugger is active!
2026-04-09T14:19:34.531597665Z * Debugger PIN: 111-527-213
2026-04-09T14:19:40.767577014Z 172.17.0.1 - - [09/Apr/2026 14:19:40] "GET /config HTTP/1.1" 200 -
```

```
# docker logs -f --tail 50 -t continued
# timestamped startup logs and HTTP access logs showing
```

Lab 4: Container Inspection

```
docker inspect flask1
# Full JSON output:
# "Id": "24dbc24c9c690a8398cf8c1355e371086461a1055e45ef1a29e4c6d3127ce1f4"
# "Created": "2026-03-02T14:30:42.967030997Z"
# "State": {
#   "Status": "running",
#   "Running": true,
#   "Paused": false,
#   "Pid": 3944,
#   "ExitCode": 0
# }
# "Name": "/monitor-test"
# "Driver": "overlayfs"
```

```
abhinav_pandey@LUCIF3R:~/flask-env$ docker inspect flask1
[
  {
    "Id": "8521cbe18941a217b3b90fee3da637e32c8e4154631552150463377b2dedf8c7",
    "Created": "2026-04-09T14:19:34.169885778Z",
    "Path": "python",
    "Args": [
      "app.py"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 9434,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-04-09T14:19:34.28243978Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:e8a28bf0a6980d29e0e62cb9d434cf0fa6ac0b79f9535514b4dfe190aca6db13",
    "ResolvConfPath": "/var/lib/docker/containers/8521cbe18941a217b3b90fee3da637e32c8e4154631552150463377b2dedf8c7/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/8521cbe18941a217b3b90fee3da637e32c8e4154631552150463377b2dedf8c7/hostname",
    "HostsPath": "/var/lib/docker/containers/8521cbe18941a217b3b90fee3da637e32c8e4154631552150463377b2dedf8c7/hosts",
    "LogPath": "/var/lib/docker/containers/8521cbe18941a217b3b90fee3da637e32c8e4154631552150463377b2dedf8c7/8521cbe18941a217b3b90fee3da637e32c8e4154631552150463377b2dedf8c7-json.log",
    "Name": "/flask1",
    "RestartCount": 0,
    "Driver": "overlayfs",
    "Platform": "linux",
    "MountLabel": "",
    "ProcessLabel": "",
    "AppArmorProfile": "",
    "ExecIDs": null,
    "HostConfig": {
      "Binds": null,
      "ContainerIDFile": "",
      "LogConfig": {
        "Type": "json-file",
        "Config": {}
      },
      "NetworkMode": "bridge",
      "PortBindings": {
        "5000/tcp": [
          {
            "HostIp": "",
            "HostPort": "5001"
          }
        ]
      },
      "RestartPolicy": {
        "Name": "no",
        "MaximumRetryCount": 0
      },
      "AutoRemove": false,

```

```

docker inspect --format='{{.Config.Env}}' flask1
# [PATH=... NGINX_VERSION=1.29.5 NJS_VERSION=0.9.5 ...]
docker inspect --format='{{.HostConfig.Memory}}' flask1
# 0
docker inspect --format='{{.HostConfig.NanoCpus}}' flask1
# 0
docker events
# (waiting, Ctrl+C to exit)
docker stop flask1
docker events
# (Ctrl+C)
docker events --filter 'type=container'
# Shows docker stats table with all running containers

```

```

abhinav_pandey@LUCIF3R:~/flask-env-apps$ docker inspect --format='{{.State.Status}}' flask1
running
abhinav_pandey@LUCIF3R:~/flask-env-apps$ docker inspect --format='{{.NetworkSettings.IPAddress}}' flask1
template parsing error: template: "1:18: executing "" at <.NetworkSettings.IPAddress>: map has no entry for key "IPAddress"
abhinav_pandey@LUCIF3R:~/flask-env-apps$ docker inspect --format='{{.Config.Env}}' flask1
UNIQUE_ID=7m21-mymw ucmmv-llue nra-oci-2awm5a2a1el raine=/usr/local/bin:/usr/local/sbin:/usr/bin:/sbin:/bin LANG=C.UTF-8 GPG_KEY=E3FF2839C048B25C084DEBE9B26995E3102
5A5A9 DVTHNM VERSION=3.0.25 DVTHNM SHA256=80e07d7c0f2f0cc002432dee84d2a40dae404a99303e3f97701c10966c91834 NODE_ENV=production PORT=3000 APP_VERSION=1.0.0]
abhinav_challa@LUCIF3R:~/flask-env-apps$ docker inspect --format='{{.HostConfig.Memory}}' flask1
0
abhinav_pandey@LUCIF3R:~/flask-env-apps$ docker inspect --format='{{.HostConfig.NanoCpus}}' flask1
0

```

Lab 5: Practical Monitoring Script

```
./monitor.sh
#=== Docker Monitoring Dashboard ===
#1. Running Containers:
#NAMES          STATUS      PORTS
#flask1         Up 2 hours  0.0.0.0:5001->3000/tcp, [::]:5001->3000/tcp
#app1          Up 3 hours  80/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
#nginx-custom   Up 3 hours  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
#new-mysql      Up 3 hours  3306/tcp, 33060/tcp
#web3           Up 3 hours  80/tcp
#web2           Up 3 hours  80/tcp
#web1           Up 3 hours  80/tcp

#2. Resource Usage:
#NAME          CPU %      MEM USAGE / LIMIT    NET I/O      BLOCK I/O
#flask1        0.23%     40.43MiB / 7.615GiB   1.92kB / 777B 0B / 258kB
#app1          0.00%     18.69MiB / 7.615GiB   2.17kB / 126B  172kB / 12.3kB
#nginx-custom  0.00%     19.02MiB / 7.615GiB   3.42kB / 1.21kB 3.91MB / 4.1kB
#new-mysql     0.82%     349.2MiB / 7.615GiB   2.46kB / 126B  36.1MB / 15.1MB
#web3          0.00%     18.82MiB / 7.615GiB   2.71kB / 126B  0B / 16.4kB
#web2          0.00%     18.88MiB / 7.615GiB   2.97kB / 126B  0B / 12.3kB
#web1          0.00%     18.62MiB / 7.615GiB   3.31kB / 126B  0B / 12.3kB

#3. Recent Events:

#4. System Info:
#TYPE          TOTAL     ACTIVE    SIZE      RECLAIMABLE
#Images        5         5         3.059GB   0B (0%)
#Containers    9         7         811kB     16.38kB (2%)
#Local Volumes 4         3         210.3MB   0B (0%)
#Build Cache   13        0         1.599GB   28.67kB
```

```

abhinav_pandey@LUCIF3R:~/fLask-env-apps ./monitor.sh
=== Docker Monitoring Dashboard ===
Time: Thu Apr 9 16:46:13 UTC 2026

1. Running Containers:
NAMES          STATUS      PORTS
flask1         Up 2 hours  0.0.0.0:5001->3000/tcp, [::]:5001->3000/tcp
app1           Up 3 hours  80/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
nginx-custom   Up 3 hours  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
new-mysql      Up 3 hours  3306/tcp, 33060/tcp
web3           Up 3 hours  80/tcp
web2           Up 3 hours  80/tcp
web1           Up 3 hours  80/tcp

2. Resource Usage:
NAME          CPU %    MEM USAGE / LIMIT    NET I/O      BLOCK I/O
flask1        0.23%   40.43MiB / 7.615GiB   1.92kB / 777B  0B / 258kB
app1          0.00%   18.69MiB / 7.615GiB   2.17kB / 126B  172kB / 12.3kB
nginx-custom  0.00%   19.02MiB / 7.615GiB   3.42kB / 1.21kB 3.91MB / 4.1kB
new-mysql     0.82%   349.2MiB / 7.615GiB   2.46kB / 126B  36.1MB / 15.1MB
web3          0.00%   18.82MiB / 7.615GiB   2.71kB / 126B  0B / 16.4kB
web2          0.00%   18.88MiB / 7.615GiB   2.97kB / 126B  0B / 12.3kB
web1          0.00%   18.62MiB / 7.615GiB   3.31kB / 126B  0B / 12.3kB

3. Recent Events:

4. System Info:
TYPE          TOTAL    ACTIVE    SIZE      RECLAIMABLE
Images        5        5         3.059GB   0B (0%)
Containers    9        7         811kB     16.38kB (2%)
Local Volumes 4        3         210.3MB   0B (0%)
Build Cache   13       0         1.599GB   28.67kB

```

Part 4: Docker Networks

Lab 1: Multi-Service Setup

```

docker network ls
#NETWORK ID      NAME      DRIVER  SCOPE
#04c34ceeaedb    bridge    bridge   local
#032eba416aaf     host     host     local
#609c188dce0e     none     null     local

```

```

abhinav_pandey@LUCIF3R:~/fLask-env-apps docker network ls
NETWORK ID      NAME      DRIVER  SCOPE
04c34ceeaedb    bridge    bridge   local
032eba416aaf     host     host     local
609c188dce0e     none     null     local

```

Lab 2: Bridge Network

```
docker run -d --name web1 --network my-network nginx
# Error: container name "/web1" already in use
docker ps -a | grep web1
docker rm -f web1
docker rm -f web2
docker run -d --name web1 --network my-network nginx
# Error: network my-network not found
docker network create my-network
```

```
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker network create my-network
monocoe097c889213bbb359818232a8c62701c9c767b321cc2d25577859426
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker network inspect my-network
{
  "Name": "my-network",
  "Id": "46b5dc6ed097c889213bbb359818232a8c62701c9c767b321cc2d25577859426",
  "Created": "2626-04-09T17:25:39.551685988Z",
  "Scope": "local",
  "Driver": "bridge",
  "EnableIPv4": true,
  "EnableIPv6": false,
  "IPAM": {
    "Driver": "default",
    "Options": {},
    "Config": [
      {
        "Subnet": "172.18.0.0/16",
        "Gateway": "172.18.0.1"
      }
    ]
  },
  "Internal": false,
  "Attachable": false,
  "Ingress": false,
  "ConfigFrom": {
    "Network": ""
  },
  "ConfigOnly": false,
  "Options": {},
  "Labels": {},
  "Containers": {},
  "Status": {
    "IPAM": {
      "Subnets": [
        {
          "172.18.0.0/16": {
            "IPsInUse": 3,
            "DynamicIPsAvailable": 65533
          }
        }
      ]
    }
  }
}
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker run -d --name web1 --network my-network nginx
docker: Error response from daemon: Conflict. The container name "/web1" is already in use by container "85ef4083cc3b5afa9dfaed7fa2426c4ab1959fdef3e6ccf06ff582da5ee88957e". You have to remove (or rename) that container to be able to reuse that name.
Run 'docker run --help' for more information
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker rm -f web1
web1
abhinav_pandey@LUCIF3R:~/flask-env-app$ docker rm -f web2
web2
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker run -d --name web1 --network my-network nginx
e9a9ab9a3abec3618d2f928e5df4825bc1cbias52420deeb49722d633d4b30713
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker run -d --name web2 --network my-network nginx
de9799af1bdc792b5d37866a3982d76ade62b072415809c8757ada8d1ae6afe
```

```
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker exec web1 curl http://web2
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 896 100 896 0 0 82855 0 --:--:-- --:--:-- --:--:-- 89600
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

```
# curl output - web1 responding with nginx default page HTML
docker run -d --name host-app --network host nginx
curl http://localhost
# curl: (7) Failed to connect to localhost port 80 after 1 ms: Connection refused
```

```
abhinav chawla@LUCIF3R:~/flask-env-app$ docker run -d --name host-app --network host nginx
bcc806882b9777fe3b0d670d21601cdf15a1b6659e09a98a53325c6b375f0ec6
abhinav_pandey@LUCIF3R:~/flask-env-app$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

Lab 3: Host & None Network

```
curl http://localhost
# Returns nginx default HTML page

docker run -d --name isolated-app --network none alpine sleep 3600
docker exec isolated-app ifconfig
# lo    Link encap:Local Loopback
#       inet addr:127.0.0.1  Mask:255.0.0.0
#       inet6 addr: ::1/128 Scope:Host
```

```

abhinav_ chawla@LUCIF3R:~/flask-env-app$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
abhinav_ chawla@LUCIF3R:~/flask-env-app$ docker run -d --name isolated-app --network none alpine sleep 3600
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
589002ba0eae: Pull complete
caa817ad3aea: Download complete
9e595aac14e0: Download complete
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
3b42026e3e065413bb811aa08e782dcf541d9c7cd14df8ba91411fed5bff9e03
abhinav_ chawla@LUCIF3R:~/flask-env-app$ docker exec isolated-app ifconfig
lo
  Link encap:Local Loopback
  inet addr:127.0.0.1  Mask:255.0.0.0
  inet6 addr: ::1/128 Scope:Host
  UP LOOPBACK RUNNING  MTU:65536  Metric:1
  RX packets:0 errors:0 dropped:0 overruns:0 frame:0
  TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
  collisions:0 txqueuelen:1000
  RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

```

```

# ifconfig continued:
# UP LOOPBACK RUNNING  MTU:65536  Metric:1
# RX packets:0 errors:0 TX packets:0 errors:0
# Only loopback - no external network access

```

```

docker network create app-network
docker network connect app-network web1
docker network disconnect app-network web1
docker network rm app-network
docker network prune
# Deleted Networks: my_overlay, my_ipvlan, 26compose_wordpress-network, mybridge

```

```

abhinav_ chawla@LUCIF3R:~/flask-env-app$ docker network create app-network
3feeac196b7ee3d735266a07529a902f74314f7f1f79fc1125461994f8205d96
abhinav_ chawla@LUCIF3R:~/flask-env-app$ docker network create --driver bridge --subnet 172.20.0.0/16 --gateway 172.20.0.1 my-subnet
76t9+2da7d8b8a4d858891df04cbae8de9f9cb6590511710f70f54efe7a6d589
abhinav_ chawla@LUCIF3R:~/flask-env-app$ docker network connect app-network web1
abhinav_ chawla@LUCIF3R:~/flask-env-app$ docker network disconnect app-network web1
abhinav_pandey@LUCIF3R:~/flask-env-app$ docker network rm app-network
app-network
abhinav_ chawla@LUCIF3R:~/flask-env-app$ docker network prune
WARNING! This will remove all custom networks not used by at least one container.
Are you sure you want to continue? [y/N] y
Deleted Networks:
my-subnet

```

Lab 4: Multi-Container App - Web + Database

```
docker network create app-network
docker run -d \
  --name postgres-db \
  --network app-network \
  -e POSTGRES_PASSWORD=secret \
  -v pgdata:/var/lib/postgresql/data \
  postgres:15
# Pulling postgres:15 - all layers Pull complete
# Status: Downloaded newer image for postgres:15
```

```
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker network create app-network
3feeac196b7ee3d735266a07529a902f74314f7f1f79fc1125461994f8205d96
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker network create --driver bridge --subnet 172.20.0.0/16 --gateway 172.20.0.1 my-subnet
7619f2da7d8b8a4d58e91df0e4bae8de9f9cbe59b511719f70f8f9e7a6d589
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker network connect app-network web1
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker network disconnect app-network web1
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker network rm app-network
app-network
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker network prune
WARNING! This will remove all custom networks not used by at least one container.
Are you sure you want to continue? [y/N] y
Deleted Networks:
my-subnet
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker network create app-network
bb849a8c65403d571ce55a97d46c98e4b58fcc6585c36b53b657a2e58c6fa93
abhinav@chawla@LUCIF3R:~/flask-env-app$ docker run -d \
  --name postgres-db \
  --network app-network \
  -e POSTGRES_PASSWORD=secret \
  -v pgdata:/var/lib/postgresql/data \
  postgres:15
Unable to find image 'postgres:15' locally
15: Pulling from library/postgres
a8a0725a32a6: Pull complete
af2a6e5f285c: Pull complete
b4766ac2e7e6: Pull complete
f7049a93e93f: Pull complete
71a32c7571f4: Pull complete
c45b346571fc: Pull complete
2c38b75ffca2: Pull complete
5ca81a59b0a4: Pull complete
9328c15dbae: Pull complete
00f33d580dc3: Pull complete
8ecd8d198da6: Pull complete
af8c3ea2fe5e: Pull complete
5c152140754e: Pull complete
b9815a4f6b87: Download complete
cebcbab07837b: Download complete
Digest: sha256:31224c68946ea75ee9522e0fd02eb202e74754adea4294fdefe9f93c875b657
Status: Downloaded newer image for postgres:15
d48c85ea9f8974b38dc24ed31fd063e25e301ceeb15913640704d52f2ea84ec
```

```
docker run -d \
  --name web-app \
  --network app-network \
  -p 8080:3000 \
  -e DATABASE_URL="postgres://postgres:secret@postgres-db:5432/mydb" \
  -e DATABASE_HOST="postgres-db" \
  node-app
# Error: pull access denied for node-app (image doesn't exist locally)

docker network inspect bridge
# "Name": "bridge"
# "Driver": "bridge"
# "EnableIPv4": true
# "IPAM": { "Config": [{ "Subnet": "172.17.0.0/16" }] }
```

```

abhinav_pandey@LUCIF3R:~/flask-env-apps docker run -d \
--name web-app \
--network app-network \
-p 8080:3000 \
-e DATABASE_URL="postgres://postgres:secret@postgres-db:5432/mydb" \
-e DATABASE_HOST="postgres-db" \
node-app
Unable to find image 'node-app:latest' locally
docker: Error response from daemon: pull access denied for node-app, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
abhinav_pandey@LUCIF3R:~/flask-env-apps docker network inspect bridge
{
  "Name": "bridge",
  "Id": "04c34ceeaedb706081edb3c302f46f3273fe40fa65434f49d8a0d7149ab61bbb",
  "Created": "2026-04-09T13:07:53.676393869Z",
  "Scope": "local",
  "Driver": "bridge",
  "EnableIPv4": true,
  "EnableIPv6": false,
  "IPAM": {
    "Driver": "default",
    "Options": null,
    "Config": [
      {
        "Subnet": "172.17.0.0/16",
        "Gateway": "172.17.0.1"
      }
    ]
  }
}

```

```

# bridge inspect continued:
# IPv4Address: "172.17.0.8/16"
docker stop $(docker ps -q)
docker rm $(docker ps -aq)
# All container IDs listed and removed

```

```

abhinav_pandey@LUCIF3R:~/flask-env-apps docker stop $(docker ps -q)
448c00eayr00
docker rm $(docker ps -aq)3b42026e3e06
bcc806882b97
de979df15bdc
e9a9ab9a3abe
8521cbe18941
45c915b7dacb
d4252783d510
9dd09592720c
d728b47aab0f
abhinav_pandey@LUCIF3R:~/flask-env-apps $(docker ps -aq)
448c00eayr00
3b42026e3e06
bcc806882b97
de979df15bdc
e9a9ab9a3abe
8521cbe18941
45c915b7dacb
d4252783d510
9dd09592720c
d728b47aab0f
ecbd2981748c
4afd69c3b1ab

```

Lab 5: Port Publishing

```
docker ps
# (empty)
docker run -d -p 80:8080 --name app1 nginx
docker run -d -p 8080 --name app2 nginx
docker ps
# app2: 0.0.0.0:56066->8080/tcp (random port assigned)
# app1: 0.0.0.0:80->8080/tcp
docker run -d -p 8082:80 -p 8443:443 --name app3 nginx
docker run -d -p 127.0.0.1:8085:80 --name app4 nginx
docker network create myapp-network
docker network ls
```

#NETWORK ID	NAME	DRIVER	SCOPE
#bb849a0c6540	app-network	bridge	local
#04c34ceeaedb	bridge	bridge	local
#032eba416aaf	host	host	local
#46b5dc6ed097	my-network	bridge	local
#5ed511317523	myapp-network	bridge	local
#609c188dce0e	none	null	local

```
abhinav_pandey@LUCIF3R:~/flask-env$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS          NAMES
abhinav_pandey@LUCIF3R:~/flask-env$ docker run -d -p 80:8080 --name app1 nginx
216d15b176b7  nginx    "/docker-entrypoint..." 4 seconds ago Up 3 seconds 80/tcp, 0.0.0.0:32768->8080/tcp, [::]:32768->8080/tcp app2
abhinav_pandey@LUCIF3R:~/flask-env$ docker run -d -p 8080 --name app2 nginx
8d603b7e1478 5e52938409b5f80b88f1e7765b47ba427242e61aea6bd21de84
abhinav_pandey@LUCIF3R:~/flask-env$ docker run -d -p 8082:80 -p 8443:443 --name app3 nginx
d8648c4982dbad4ce496baaeca2a61641b62dc456d48277e7d93bc8b5f7d8256
abhinav_pandey@LUCIF3R:~/flask-env$ docker run -d -p 127.0.0.1:8085:80 --name app4 nginx
d8fe0179fc5e63bd19761580bbe137b649b91a16747aeb246b9c7b29ba264a02
abhinav_pandey@LUCIF3R:~/flask-env$ docker network create myapp-network
5ed51131752394ca52548188ea60d66b12e559600828cbaac8279e4dce4d9d21
abhinav_pandey@LUCIF3R:~/flask-env$ docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
bb849a0c6540	app-network	bridge	local
04c34ceeaedb	bridge	bridge	local
032eba416aaf	host	host	local
46b5dc6ed097	my-network	bridge	local
5ed511317523	myapp-network	bridge	local
609c188dce0e	none	null	local

Part 5: Complete Real-World Example

Flask + PostgreSQL + Redis Setup

```
docker run -d \
  --name redis \
  --network myapp-network \
  -v redis-data:/data \
  redis:7-alpine
# Pulling redis:7-alpine - all layers Pull complete
# Status: Downloaded newer image for redis:7-alpine
mkdir app
echo "print('Flask app placeholder')" > app/app.py
echo "SECRET_KEY=supersecretkey" > .env.production
echo "APP_ENV=production" >> .env.production
cat .env.production
# SECRET_KEY=supersecretkey
# APP_ENV=production
docker build -t flask-app:latest .
# [+] Building 61.0s (7/7) FINISHED
```

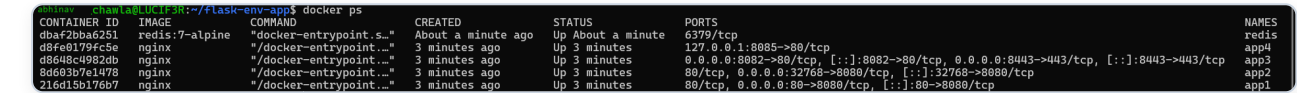
```

abhinav_pandey@LUCIF3R:~/flask-env-apps docker run -d \
--name flask_app \
--network myapp-network \
-v redis-data:/data \
redis:7-alpine
Unable to find image 'redis:7-alpine' locally
7-alpine: Pulling from library/redis
9ef37f0a2ac: Pull complete
1c28d8ee428: Pull complete
09a5a0c32a23: Pull complete
bc1da058f299: Pull complete
2a9e5d33d69a: Pull complete
bd53938b1271: Pull complete
7ad54b3c4cef: Pull complete
444fb708ef50: Pull complete
9bbcfac5d580: Download complete
06c2db80d40e: Download complete
Digest: sha256:8b81dd37ff027bec4e156d41acfbe92460870dc6444579a2ac5b9d59df0665
Status: Downloaded newer image for redis:7-alpine
d0af2bba251de9d53a34559fe80737f4e923d7f565e0eb190e08f9493ec5c
abhinav_pandey@LUCIF3R:~/flask-env-apps$ docker exec -it flask_app /bin/bash
abhinav_challa@LUCIF3R:~/flask-env-app$ SECRET_KEY=supersecretkey >> .env.production
echo "APP_ENV=production" >> .env.production
abhinav_pandey@LUCIF3R:~/flask-env-app$ flask env update --prod
SECRET_KEY=supersecretkey
APP_ENV=production
abhinav_pandey@LUCIF3R:~/flask-env-app$ docker build -t flask-app:latest .
[+] Building 44.5s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 300B
=> [internal] load metadata for docker.io/library/python:3.9
=> [internal] load dockerimage python:3.9
=> transferring context: 2B
[1/4] FROM docker.io/library/python:3.9@sha256:da5ae29682d12a6649f5fc18d6f15b7874eb3efc524b923cd1d8cb3380d07c913
=> resolve docker.io/library/python:3.9@sha256:da5ae29682d12a6649f5fc18d6f15b7874eb3efc524b923cd1d8cb3380d07c913
=> [internal] load build context
=> transferring context: 796B
=> CACHED [2/4] WORKDIR /app
=> [3/4] COPY .
=> [4/4] RUN pip install flask
=> exporting to image
=> exporting layers
=> exporting manifest sha256:f8478b10932bc3582891c8dce45aac4efa7dfbf6be0e074fd06ca20e2164c990
=> exporting config sha256:34262fa4e4ff4d499eeaa6d65df6d9a5841b19d14814cc7da19351a6557787
=> exporting attestation manifest sha256:f2b8a37bcsd1c15e999ff39803af69a9d839dc7344b27f1cd2f77de98972b
=> exporting manifest list sha256:e46fcca91464f6f95a92d6938f3dc2ba33d7a9f86d04eed1c48f5f0b45bee29
=> naming to docker.io/library/flask-app:latest
=> unpacking to docker.io/library/flask-app:latest

```

Monitoring All Services

```
# docker stats (no-stream) - postgres, redis, flask-app all showing --
# Ctrl+C
docker ps
#CONTAINER ID      IMAGE               COMMAND              CREATED             STATUS
#dbaf2bba6251      redis:7-alpine     "docker-entrypoint.s..." About a minute ago  Up About a minute
#d8fe0179fc5e      nginx              "/docker-entrypoint...." 3 minutes ago      Up 3 minutes
#d8648c4982db      nginx              "/docker-entrypoint...." 3 minutes ago      Up 3 minutes
#8d603b7e1478      nginx              "/docker-entrypoint...." 3 minutes ago      Up 3 minutes
#216d15b176b7      nginx              "/docker-entrypoint...." 3 minutes ago      Up 3 minutes
```



CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
dbaf2bba6251	redis:7-alpine	"docker-entrypoint.s..."	About a minute ago	Up About a minute	6379/tcp	redis
d8fe0179fc5e	nginx	"/docker-entrypoint...."	3 minutes ago	Up 3 minutes	127.0.0.1:8085->80/tcp	app4
d8648c4982db	nginx	"/docker-entrypoint...."	3 minutes ago	Up 3 minutes	0.0.0.0:8082->80/tcp, [::]:8082->80/tcp, 0.0.0.0:8443->443/tcp, [::]:8443->443/tcp	app3
8d603b7e1478	nginx	"/docker-entrypoint...."	3 minutes ago	Up 3 minutes	80/tcp, 0.0.0.0:32768->8080/tcp, [::]:32768->8080/tcp	app2
216d15b176b7	nginx	"/docker-entrypoint...."	3 minutes ago	Up 3 minutes	80/tcp, 0.0.0.0:80->8080/tcp, [::]:80->8080/tcp	app1

Key Takeaways

1. **Volumes** persist data beyond container lifecycle
2. **Environment variables** configure containers dynamically
3. **Monitoring commands** help debug and optimize containers
4. **Networks** enable secure container communication
5. **Always use named volumes** for production data
6. **Custom networks** provide better isolation and DNS
7. **Monitor resource usage** to prevent issues
8. **Use .env files** for sensitive configuration